GOLANG MCQs

Module 1 & 2: Go Basics and Foundations

Q1. What is the entry point of a Go program?
A. start()
B. init()
C. main()
D. run()
Answer: C

Q2. Which keyword is used to declare a variable in Go?
A. let
B. define
C. var
D. int
Answer: C

Q3. Which package is commonly used for formatted output in Go?
A. io
B. fmt
C. os
D. bufio
Answer: B

Q4. How do you declare a constant in Go?
A. final
B. static
C. let
D. const
Answer: D

Q5. Which data type is used for whole numbers in Go?
A. float
B. string
C. bool
D. int
Answer: D

Q6. Which keyword is used to create loops in Go?
A. while
B. loop
C. do
D. for
Answer: D

Q7. Which keyword is used to define a function in Go?
A. fn
B. func
C. function
D. def
Answer: B

Q8. Which operator is used to get the address of a variable in Go?
A. *
B. %
C. &
D. #
Answer: C

Q9. Which function is commonly used to create a slice in Go?
A. new()
B. make()
C. build()
D. create()
Answer: B

Q10. Which data structure stores key-value pairs in Go?
A. array
B. slice
C. map
D. struct
Answer: C

Q11. How are errors typically handled in Go?
A. try-catch
B. throw
C. catch

D. if err != nil
Answer: D

Q12. What is the main purpose of defer in Go?
A. Stop execution
B. Delay execution until function ends
C. Execute loop faster
D. Create goroutines
Answer: B

Q13. What will be the output?

```
package main
import "fmt"

func main() {
   for i := 1; i <= 3; i++ {
      fmt.Print(i)
   }
}
```

A. 123
B. 321
C. 0123
D. Compilation error
Answer: A

Q14. What will be the output?

```
x := 2

switch x {
case 1:
   fmt.Println("One")
case 2:
   fmt.Println("Two")
default:
   fmt.Println("Other")
```

}

A. One
B. Two
C. Other
D. Compilation error
Answer: B

Q15. What happens when a matching case is found in Go switch statement?
A. All remaining cases execute
B. Execution stops automatically
C. break is mandatory
D. default always executes
Answer: B

Q16. What will be the output?

```
func main() {
   defer fmt.Println("First")
   defer fmt.Println("Second")
}
```

A. First Second
B. Second First
C. First First
D. No Output
Answer: B

Q17. In the function call greet("Go"), what is "Go"?
A. Parameter
B. Return value
C. Argument
D. Function body
Answer: C

Q18. Which is a valid function definition in Go?
A. function add(a int) {}
B. func add(a, b int) {}
C. func add(int a) {}
D. define add() {}
Answer: B

Q19. What does the following code do?

```
type Age int
```

A. Creates variable Age
B. Creates custom type Age
C. Imports Age package
D. Converts int to string
Answer: B

Q20. Which is the correct pointer declaration?
A. var p int*
B. var *p int
C. var p *int
D. int *p
Answer: C

Module 3: Data Structures

Q21. Which function is used to get the number of elements in an array?
A. size()
B. count()
C. len()
D. length()
Answer: C

Q22. What will be the output?

```
arr := [3]int{10,20,30}
```

```
fmt.Println(arr[1])
```

A. 10
B. 20
C. 30
D. Error
Answer: B

Q23. What will be the output?

```
nums := []int{1,2,3}
fmt.Println(len(nums), cap(nums))
```

A. 3 3
B. 0 3
C. 3 0
D. Compilation error
Answer: A

Q24. What does append() function do?
A. Removes elements
B. Adds elements to slice
C. Sorts slice
D. Changes only capacity
Answer: B

Q25. What is the purpose of range keyword in Go?
A. Create arrays
B. Compare values
C. Loop through collections
D. Declare variables
Answer: C

Q26. What happens when range reaches the last element?
A. Infinite loop
B. Crash
C. Manual break needed

D. Loop stops automatically
Answer: D

Q27. Which is a valid map declaration in Go?
A. marks := map[string]int{"A":90}
B. marks = ["A":90]
C. map()
D. create(map)
Answer: A

Q28. What happens if the same key is added again in a map?
A. Error
B. Old value replaced
C. Both values stored
D. Map resets
Answer: B

Q29. Which keyword is used to define an interface in Go?
A. struct
B. interface
C. type
D. implement
Answer: B

Q30. What will be the output?

```
type Printer interface {
   Print()
}

type Hello struct {}

func (h Hello) Print() {
   fmt.Println("Hello")
}

func main() {
   var p Printer
   p = Hello{}
```

```
    p.Print()
}
```

A. Hello
B. Runtime error
C. Compile error
D. Nothing
Answer: A

Module 4: Concurrency

Q31. Which keyword is used to start a goroutine?
A. async
B. thread
C. go
D. start
Answer: C

Q32. What is the primary purpose of goroutines?
A. Replace functions
B. Execute tasks concurrently
C. Improve syntax
D. Replace loops
Answer: B

Q33. What is the primary role of Go scheduler?
A. Database handling
B. Manage goroutines execution
C. Handle HTTP requests
D. Manage files
Answer: B

Q34. How does Go scheduler improve performance?
A. Increases CPU speed
B. Runs many goroutines on few threads
C. Removes memory usage

D. Executes all simultaneously
Answer: B

Q35. What is a channel mainly used for?
A. File handling
B. Data sharing between goroutines
C. Logging
D. HTTP routing
Answer: B

Q36. What does this code do?

```
msg := <-ch
```

A. Sends data
B. Creates channel
C. Receives data
D. Closes channel
Answer: C

Q37. What is the main purpose of select statement in Go?
A. Create goroutines
B. Close channels
C. Choose between channel operations
D. Declare channels
Answer: C

Q38. When does default case run in select?
A. Always
B. Program start
C. When no channel is ready
D. Channel closed
Answer: C

Q39. What is the purpose of context package?
A. Create arrays
B. Handle cancellation and deadlines
C. Format strings

D. Create APIs
Answer: B

Q40. Which package provides Mutex and WaitGroup?
A. net/http
B. context
C. sync
D. time
Answer: C

Q41. What is the purpose of WaitGroup?
A. Share data
B. Wait for goroutines to finish
C. Pause threads
D. Lock channels
Answer: B

Q42. Which method increases WaitGroup counter?
A. wg.Done()
B. wg.Wait()
C. wg.Add()
D. wg.Start()
Answer: C

Q43. Which situation causes race condition?
A. Multiple goroutines modifying same variable
B. Using defer
C. Using maps
D. Using fmt package
Answer: A

Q44. What does Mutex do?
A. Stops goroutines permanently
B. Waits for completion
C. Allows only one goroutine to access shared data
D. Creates goroutines
Answer: C

Q45. Which line causes deadlock?

```
mu.Lock()
mu.Lock()
```

A. First Lock()
B. Second Lock()
C. Both lines
D. No deadlock
Answer: B

Q46. Which is the safest way to unlock mutex?
A. Unlock manually later
B. Never unlock
C. defer mu.Unlock()
D. Use sleep
Answer: C

Q47. Concurrency means:
A. Multiple CPUs only
B. Multiple tasks progressing together
C. Faster clock speed
D. One task only
Answer: B

Q48. Parallelism means:
A. Tasks running simultaneously on multiple CPUs
B. Single task execution
C. Using loops
D. Memory optimization
Answer: A

Q49. Which package is commonly used for time delays?
A. delay
B. timer
C. sync
D. time
Answer: D

Q50. What will be printed first?

```
go fmt.Println("A")
fmt.Println("B")
```

A. Always A
B. Always B
C. A then B
D. Unpredictable
Answer: D

Module 5: Advanced Go

Q51. What are generics mainly used for?
A. File handling
B. Writing reusable code
C. Concurrency only
D. JSON encoding
Answer: B

Q52. Which symbol is used in generic type parameters?
A. {}
B. []
C. <>
D. ()
Answer: C

Q53. Which file manages Go module dependencies?
A. package.json
B. go.mod
C. requirements.txt
D. pom.xml
Answer: B

Q54. Which command initializes a Go module?
A. go init
B. go start
C. go mod init

D. go build init
Answer: C

Q55. Which command downloads dependencies in Go?
A. go install
B. go fetch
C. go mod tidy
D. go dep
Answer: C

Q56. What is the purpose of packages in Go?
A. Increase CPU usage
B. Organize reusable code
C. Create databases
D. Run goroutines
Answer: B

Q57. Which keyword imports packages?
A. using
B. require
C. include
D. import
Answer: D

Q58. Which package is used for formatted input/output?
A. net/http
B. fmt
C. sync
D. context
Answer: B

Q59. What does go build do?
A. Deletes code
B. Compiles Go program
C. Creates API
D. Starts server only
Answer: B

Q60. Which Go command runs a program directly?
A. go execute
B. go launch
C. go run
D. go compile
Answer: C

REST APIs

Q61. What best describes an API?
A. Stores data permanently
B. Creates UI
C. Takes request and returns response
D. Browser-only tool
Answer: C

Q62. Which HTTP method fetches data without changing it?
A. POST
B. DELETE
C. PUT
D. GET
Answer: D

Q63. What does json.NewEncoder(w).Encode(product) do?
A. Reads JSON
B. Converts struct to JSON and sends response
C. Saves to DB
D. Creates router
Answer: B

Q64. What is the main role of a router in REST API?
A. Store data
B. Connect DB
C. Decide which handler should run
D. Convert JSON
Answer: C

Q65. Router decides handler based on:
A. Status code
B. JSON structure
C. URL path and HTTP method
D. Request body only
Answer: C

Q66. Middleware is mainly used for:
A. Authentication and logging
B. Storing files
C. Database replication
D. Image compression
Answer: A

Q67. Which package is commonly used to build HTTP servers in Go?
A. io/http
B. http
C. net/http
D. server
Answer: C

Q68. REST stands for:
A. Remote Execution Standard Transfer
B. Representational State Transfer
C. Request State Transfer
D. Reliable State Transport
Answer: B

Q69. gRPC is mainly based on:
A. XML
B. JSON
C. Protocol Buffers
D. HTML
Answer: C

Q70. REST APIs commonly use which data format?
A. CSV
B. XML only

C. Binary only
D. JSON
Answer: D

SYSTEM DESIGN MCQs

Q71. Scalability means:
A. Faster UI colors
B. Ability to handle growing traffic
C. Database deletion
D. Low memory usage only
Answer: B

Q72. Availability means:
A. System is accessible when needed
B. Faster code compilation
C. High CPU usage
D. Using CDN
Answer: A

Q73. Reliability mainly focuses on:
A. System consistency and correctness
B. UI design
C. Image rendering
D. Code comments
Answer: A

Q74. Latency refers to:
A. Number of users
B. Time taken to process request
C. Storage size
D. Server cost
Answer: B

Q75. Throughput means:
A. Requests handled per second
B. Disk space

C. CPU speed
D. Network cable length
Answer: A

Q76. In CAP theorem, C stands for:
A. Cache
B. Consistency
C. Cluster
D. Communication
Answer: B

Q77. Which component distributes traffic across servers?
A. CDN
B. Reverse Proxy
C. Load Balancer
D. Cache
Answer: C

Q78. Reverse proxy sits:
A. Behind database
B. Between clients and servers
C. Inside browser
D. In operating system
Answer: B

Q79. Which database scaling method splits data across servers?
A. Replication
B. Caching
C. Sharding
D. Backup
Answer: C

Q80. Replication mainly improves:
A. Availability and read performance
B. UI
C. CPU frequency
D. Compilation speed
Answer: A

Q81. Message queues are mainly used for:
A. Blocking requests
B. Asynchronous processing
C. Image editing
D. SQL joins
Answer: B

Q82. Which caching strategy removes least recently used data?
A. FIFO
B. Random
C. LRU
D. Round Robin
Answer: C

Q83. CDN mainly helps reduce:
A. Server RAM
B. Latency for global users
C. Database schema
D. API endpoints
Answer: B

Q84. Which is an example of trade-off in system design?
A. Consistency vs Availability
B. Variables vs Constants
C. Arrays vs Loops
D. JSON vs HTML
Answer: A

Q85. Which database is generally preferred for highly relational data?
A. NoSQL
B. Graph DB
C. SQL
D. Cache DB
Answer: C

DSA MCQs

Q86. Which traversal uses queue?
A. DFS
B. BFS
C. Inorder
D. Topological sort
Answer: B

Q87. Which traversal uses stack/recursion?
A. BFS
B. DFS
C. Level order
D. Heap traversal
Answer: B

Q88. Topological sorting is applicable on:
A. Undirected graph
B. Weighted graph
C. DAG
D. Tree only
Answer: C

Q89. Which algorithm finds shortest path in weighted graph with non-negative weights?
A. Prim's
B. Kruskal's
C. Dijkstra
D. DFS
Answer: C

Q90. Which data structure is commonly used in Dijkstra algorithm?
A. Queue
B. Stack
C. Priority Queue
D. Linked List
Answer: C

Q91. Which algorithm is used to find Minimum Spanning Tree?
A. BFS
B. DFS

C. Kruskal
D. Binary Search
Answer: C

Q92. DSU stands for:
A. Dynamic Set Utility
B. Disjoint Set Union
C. Data Structure Unit
D. Distributed Set Utility
Answer: B

Q93. Which greedy algorithm selects activities based on finishing time?
A. Huffman Coding
B. Dijkstra
C. Activity Selection
D. LIS
Answer: C

Q94. Huffman Coding is mainly used for:
A. Sorting
B. Compression
C. Graph traversal
D. Searching
Answer: B

Q95. Fractional Knapsack uses:
A. Backtracking
B. Dynamic Programming
C. Greedy approach
D. DFS
Answer: C

Q96. Dynamic Programming mainly avoids:
A. Variables
B. Recursion
C. Repeated calculations
D. Arrays
Answer: C

Q97. Memoization is:
A. Bottom-up approach
B. Top-down with caching
C. Sorting method
D. Graph traversal
Answer: B

Q98. Tabulation is:
A. Recursive approach
B. Bottom-up DP approach
C. Greedy technique
D. BFS method
Answer: B

Q99. Which DP problem finds longest increasing subsequence?
A. Knapsack
B. Coin Change
C. LIS
D. MST
Answer: C

Q100. In Coin Change problem, DP state usually represents:
A. Number of graphs
B. Minimum coins for amount
C. Queue size
D. Tree height
Answer: B

Additional 20 DSA MCQs (Complexity + Advanced Concepts)

Q101. What is the time complexity of BFS traversal in a graph?
A. O(V)
B. O(E)
C. O(V + E)
D. $O(V^2)$
Answer: C

Q102. What is the space complexity of BFS?
A. O(1)

B. O(V)
C. O(E)
D. O(log V)
Answer: B

Q103. What is the time complexity of DFS traversal?
A. O(V + E)
B. $O(V^2)$
C. $O(E^2)$
D. O(log V)
Answer: A

Q104. Which data structure is commonly used in DFS implementation?
A. Queue
B. Heap
C. Stack
D. HashMap
Answer: C

Q105. What is the time complexity of Dijkstra Algorithm using Priority Queue?
A. $O(V^2)$
B. O(E log V)
C. O(V log E)
D. $O(E^2)$
Answer: B

Q106. Which shortest path algorithm works with negative edge weights?
A. Dijkstra
B. Prim's
C. Bellman-Ford
D. BFS
Answer: C

Q107. What is the time complexity of Kruskal's Algorithm?
A. O(E log E)
B. $O(V^2)$

C. O(V + E)
D. O(log V)
Answer: A

Q108. Which data structure is mainly used in Kruskal's Algorithm?
A. Stack
B. Queue
C. DSU / Union Find
D. Linked List
Answer: C

Q109. What is the average time complexity of Union-Find with path compression?
A. O(n)
B. O(log n)
C. Nearly O(1)
D. O(n^2)
Answer: C

Q110. Which algorithm uses greedy strategy?
A. Merge Sort
B. Huffman Coding
C. Binary Search
D. DFS
Answer: B

Q111. What is the time complexity of Activity Selection Algorithm after sorting?
A. O(n)
B. O(log n)
C. O(n log n)
D. O(n^2)
Answer: C

Q112. What is the time complexity of Fractional Knapsack Algorithm?
A. O(n^2)
B. O(n log n)
C. O(log n)

D. O(V+E)
Answer: B

Q113. Which approach is mainly used in Dynamic Programming?
A. Divide and Conquer + Storing Results
B. Greedy only
C. Backtracking only
D. Randomization
Answer: A

Q114. What is the main advantage of Memoization?
A. Reduces recursion calls
B. Removes arrays
C. Improves sorting speed
D. Removes loops
Answer: A

Q115. Which DP approach fills table iteratively?
A. Memoization
B. Recursion
C. Tabulation
D. DFS
Answer: C

Q116. What is the time complexity of 0/1 Knapsack DP solution?
A. O(nW)
B. $O(n^2)$
C. O(log n)
D. $O(W^2)$
Answer: A

Q117. What is the space complexity of classic 0/1 Knapsack DP table?
A. O(1)
B. O(W)
C. O(nW)
D. O(log n)
Answer: C

Q118. What is the optimized space complexity of Knapsack using space optimization?
A. $O(n^2)$
B. O(W)
C. O(log W)
D. O(nW)
Answer: B

Q119. What is the time complexity of Longest Increasing Subsequence using DP?
A. O(n)
B. O(n log n)
C. $O(n^2)$
D. O(log n)
Answer: C

Q120. Which algorithmic paradigm is used in Coin Change problem?
A. Greedy
B. Divide and Conquer
C. Dynamic Programming
D. Graph Traversal
Answer: C